

```

import java.util.ArrayList;
import java.util.Scanner;

/**
 * CodeAlpha Java Internship - Task 1
 * Student Grade Tracker
 *
 * Features:
 * - Add students
 * - Add grades for a student
 * - View individual student report
  (average, highest, lowest)
 * - View summary report of ALL students
 * - Uses ArrayList to store dynamic data
 */
public class StudentGradeTracker {

    // ----- Student class -----
    static class Student {
        private String name;
        private ArrayList<Double> grades;

        public Student(String name) {
            this.name = name;
            this.grades = new ArrayList<>
();
        }

        public void addGrade(double grade)
{

```

```

        grades.add(grade);
    }

    public String getName() {
        return name;
    }

    public ArrayList<Double>
getGrades() {
        return grades;
    }

    public double getAverage() {
        if (grades.isEmpty()) return
0.0;

        double sum = 0;
        for (double g : grades) sum +=
g;

        return sum / grades.size();
    }

    public double getHighest() {
        if (grades.isEmpty()) return
0.0;

        double max = grades.get(0);
        for (double g : grades) if (g >
max) max = g;
        return max;
    }

    public double getLowest() {
        if (grades.isEmpty()) return
0.0;

        double min = grades.get(0);

```

```

        for (double g : grades) if (g <
min) min = g;
        return min;
    }
}

// ----- Main program -----
private static ArrayList<Student>
students = new ArrayList<>();
private static Scanner sc = new
Scanner(System.in);

public static void main(String[] args)
{

System.out.println("=====
=====");
        System.out.println("    STUDENT
GRADE TRACKER - CodeAlpha ");

System.out.println("=====
=====");

        boolean running = true;
        while (running) {
            printMenu();
            String choice =
sc.nextLine().trim();

            switch (choice) {
                case "1":
                    addStudent();
                    break;
                case "2":

```

```

        addGradeToStudent();
        break;
    case "3":
        viewStudentReport();
        break;
    case "4":
        viewSummaryReport();
        break;
    case "5":
        running = false;

System.out.println("Exiting... Goodbye!");
        break;
    default:

System.out.println("Invalid choice! Please
try again.\n");
    }
}
sc.close();
}

private static void printMenu() {
    System.out.println("\n-----
MENU -----");
    System.out.println("1. Add
Student");
    System.out.println("2. Add Grade to
Student");
    System.out.println("3. View
Individual Student Report");
    System.out.println("4. View Summary
Report (All Students)");
    System.out.println("5. Exit");
}

```

```

        System.out.print("Enter your
choice: ");
    }

    private static void addStudent() {
        System.out.print("Enter student
name: ");
        String name = sc.nextLine().trim();
        if (name.isEmpty()) {
            System.out.println("Name cannot
be empty.");
            return;
        }
        students.add(new Student(name));
        System.out.println("Student \"\" +
name + "\" added successfully.");
    }

    private static Student
findStudent(String name) {
        for (Student s : students) {
            if
(s.getName().equalsIgnoreCase(name)) return
s;
        }
        return null;
    }

    private static void addGradeToStudent()
{
        if (students.isEmpty()) {
            System.out.println("No students
found. Please add a student first.");
            return;
        }
    }

```

```

        }
        System.out.print("Enter student
name: ");
        String name = sc.nextLine().trim();
        Student s = findStudent(name);
        if (s == null) {
            System.out.println("Student not
found.");
            return;
        }
        System.out.print("Enter grade (0-
100): ");
        try {
            double grade =
Double.parseDouble(sc.nextLine().trim());
            if (grade < 0 || grade > 100) {
                System.out.println("Grade
must be between 0 and 100.");
                return;
            }
            s.addGrade(grade);
            System.out.println("Grade added
for " + s.getName());
        } catch (NumberFormatException e) {
            System.out.println("Invalid
number entered.");
        }
    }

    private static void viewStudentReport()
{
        if (students.isEmpty()) {
            System.out.println("No students
found.");
        }
    }
}

```

```

        return;
    }
    System.out.print("Enter student
name: ");
    String name = sc.nextLine().trim();
    Student s = findStudent(name);
    if (s == null) {
        System.out.println("Student not
found.");
        return;
    }
    System.out.println("\n---- Report
for " + s.getName() + " ----");
    System.out.println("Grades    : " +
s.getGrades());
    System.out.printf("Average    :
%.2f\n", s.getAverage());
    System.out.printf("Highest    :
%.2f\n", s.getHighest());
    System.out.printf("Lowest     :
%.2f\n", s.getLowest());
    }

    private static void viewSummaryReport()
    {
        if (students.isEmpty()) {
            System.out.println("No students
found.");
            return;
        }
    }

    System.out.println("\n=====
SUMMARY REPORT =====");
    System.out.printf("%-15s %-10s

```

```
%-10s %-10s%n", "Name", "Average",  
"Highest", "Lowest");  
    System.out.println("-----  
-----");  
    for (Student s : students) {  
        System.out.printf("%-15s  
%-10.2f %-10.2f %-10.2f%n",  
            s.getName(),  
s.getAverage(), s.getHighest(),  
s.getLowest());  
    }  
  
    System.out.println("=====  
=====");  
    }  
}
```